



Washington Coffee Matcher

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Background



Seattle has 56 coffee shops per 100,000 people - over 4x more than national average.

We spend \$54 a month on coffee compared to \$40 for national average.

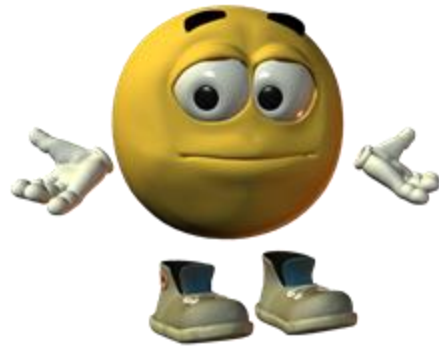
Cafe Encore was the first coffee house in Seattle, opened 1958 in the U-district.



Users & Components

Two Types of Retail Clients

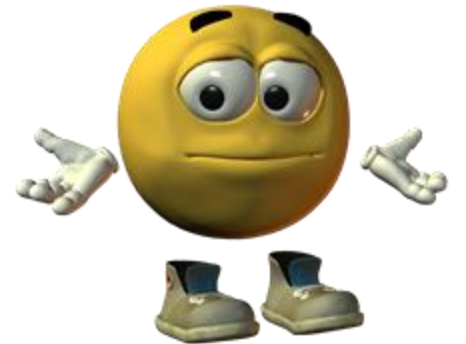
Single match?



What do I do?



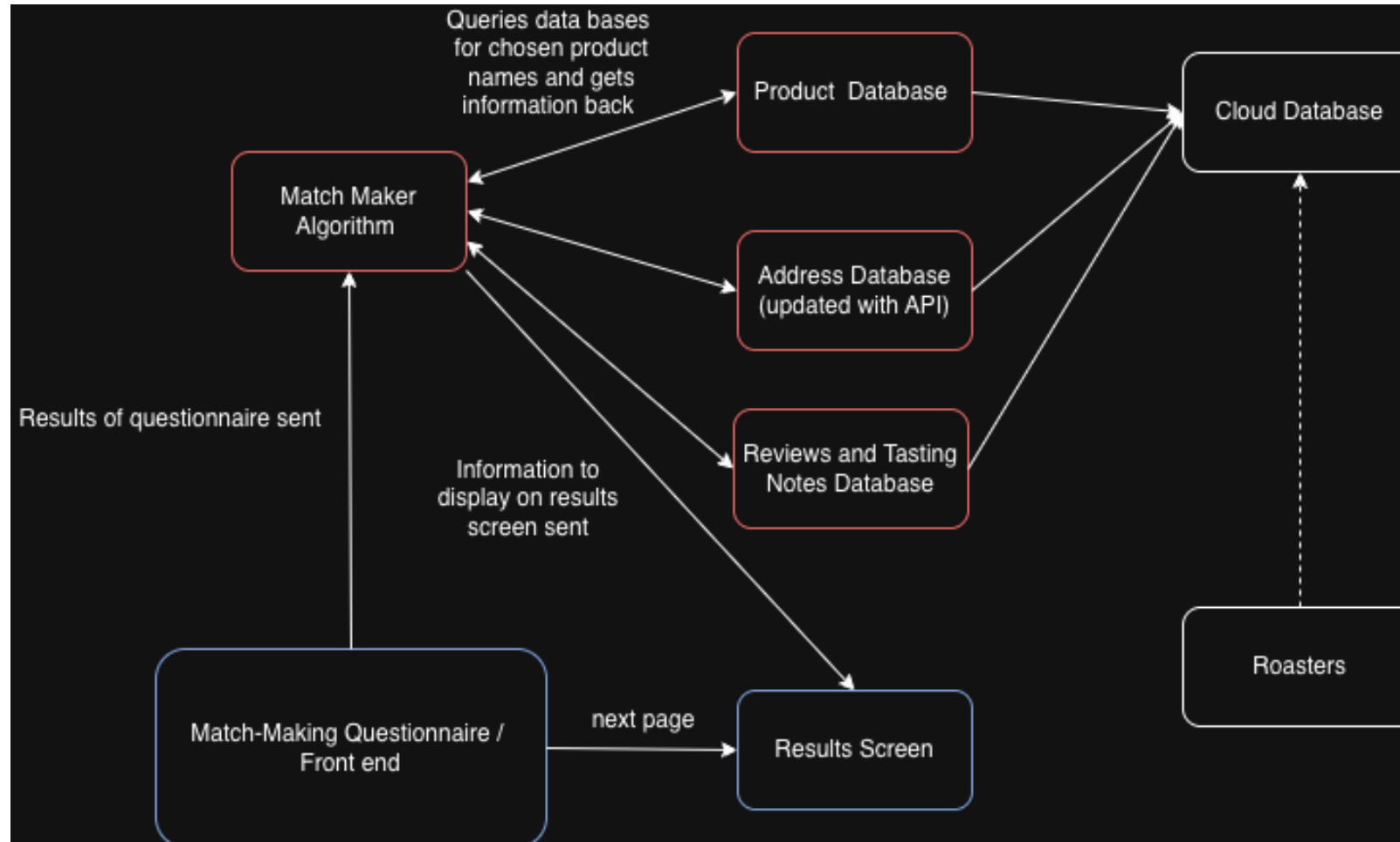
Multiple match?



So many amazing local roasters



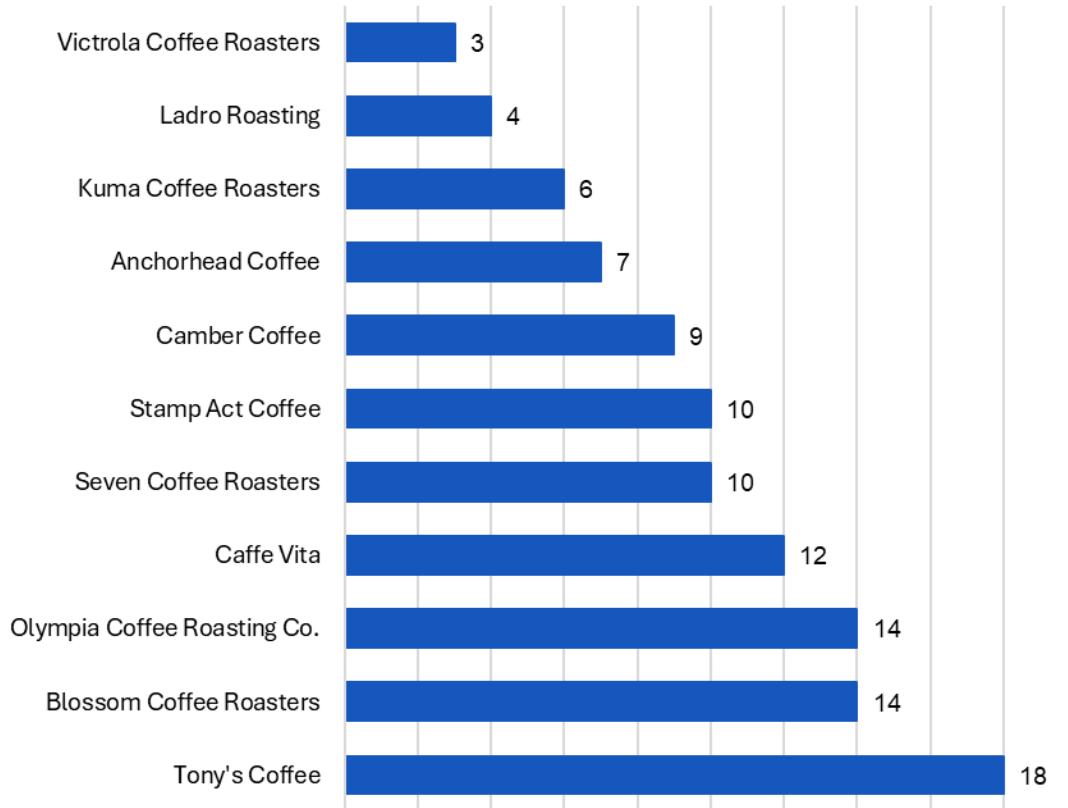
Components



Data

Data: Coffee, Class, and County

Our sample of Washington roasters and their number of coffee types (scraped data)

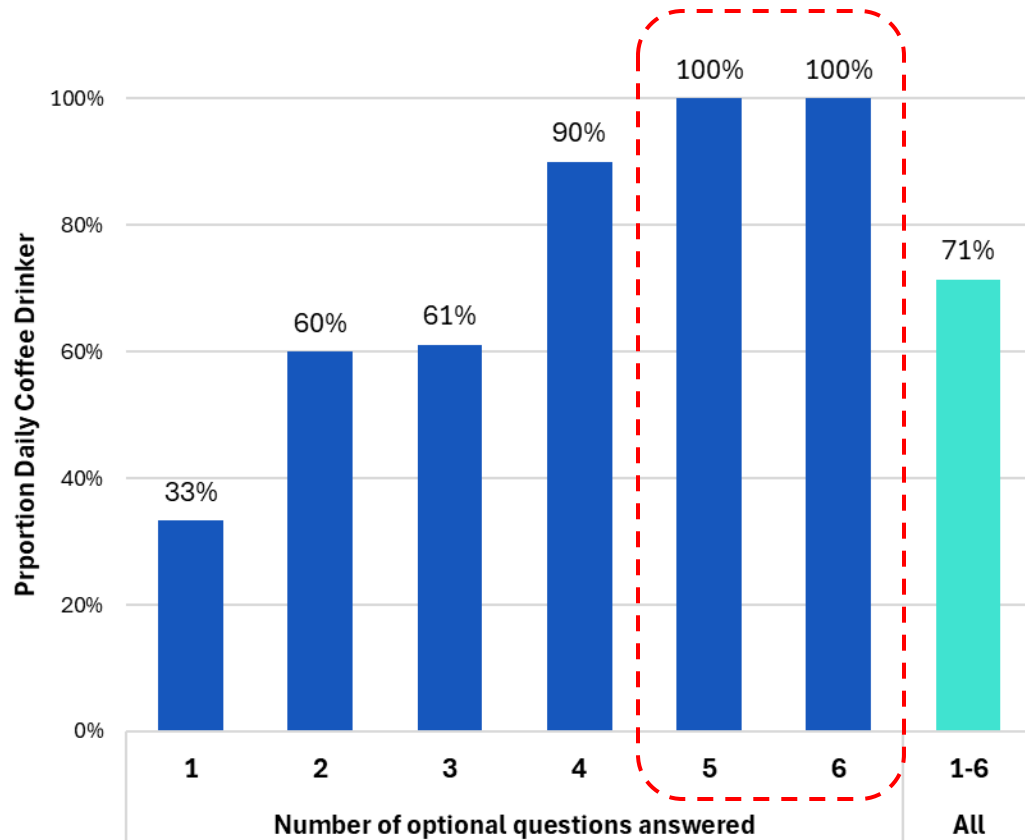


- Two primary datasets (scraped)
 - Products: Price, Shade, Origin, Rating, etc.
 - Reviews: Tasting notes, Reviews
- Supporting Class Survey
- Supporting "Business Name" data search
 - King County Restaurant Inspection API
 - 60-70% match rate
 - ~~Washington Department of Revenue~~
 - Not machine searchable
 - ~~Kroger Product API~~
 - Low match rate anticipated

User Survey

Survey and Daily Coffee

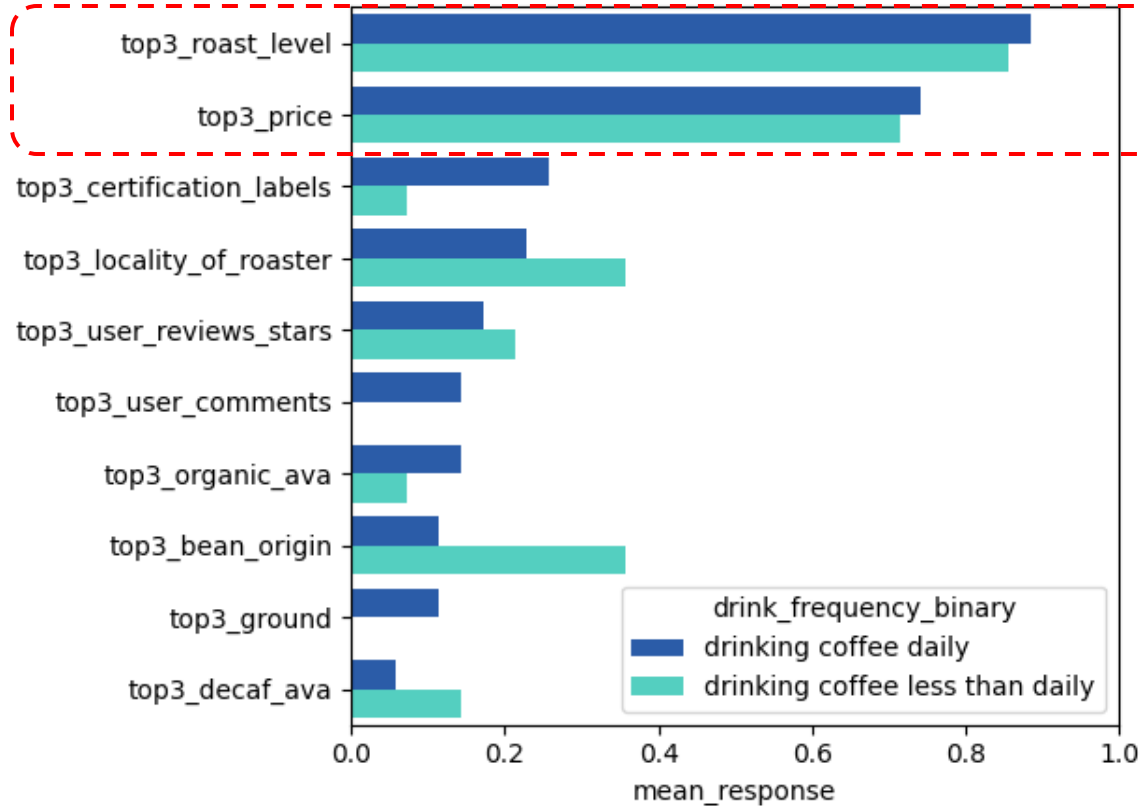
Optional questions responded to and likelihood of drinking one or more cups of coffee each day



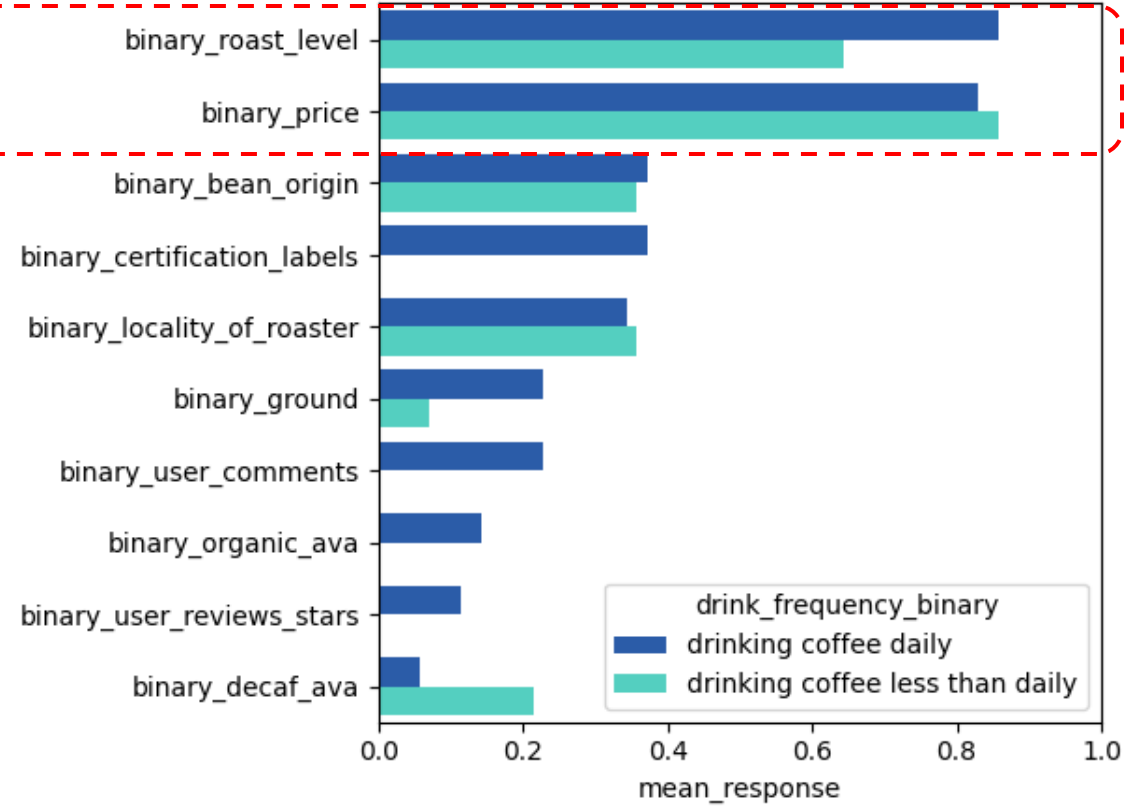
- We appreciate the surveys
- Validates avid coffee consumer's interest
- But data lacks complete coffee information

Survey Features

Factors when prioritized in top3

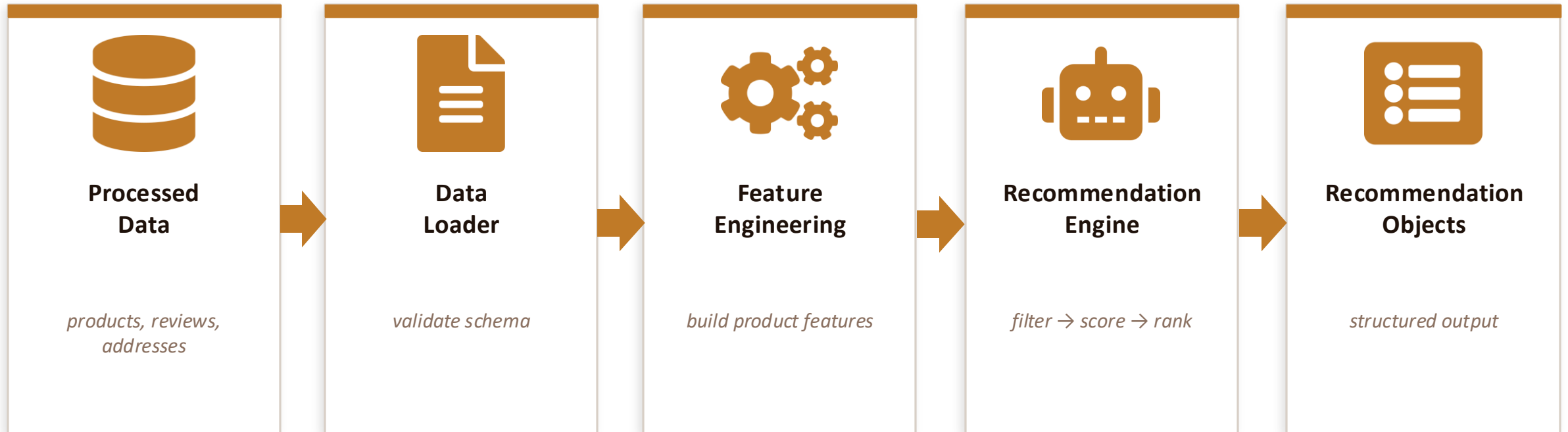


All factors available for selection



Backend Design

CoffeeMatch — Backend Architecture



Modular Recommendation Pipeline

The Bag-Size Comparison Problem



12 oz

\$18

\$1.50 / oz

Standard



24 oz

\$30

\$1.25 / oz

Mid Bulk



5 lb

\$60

\$0.75 / oz

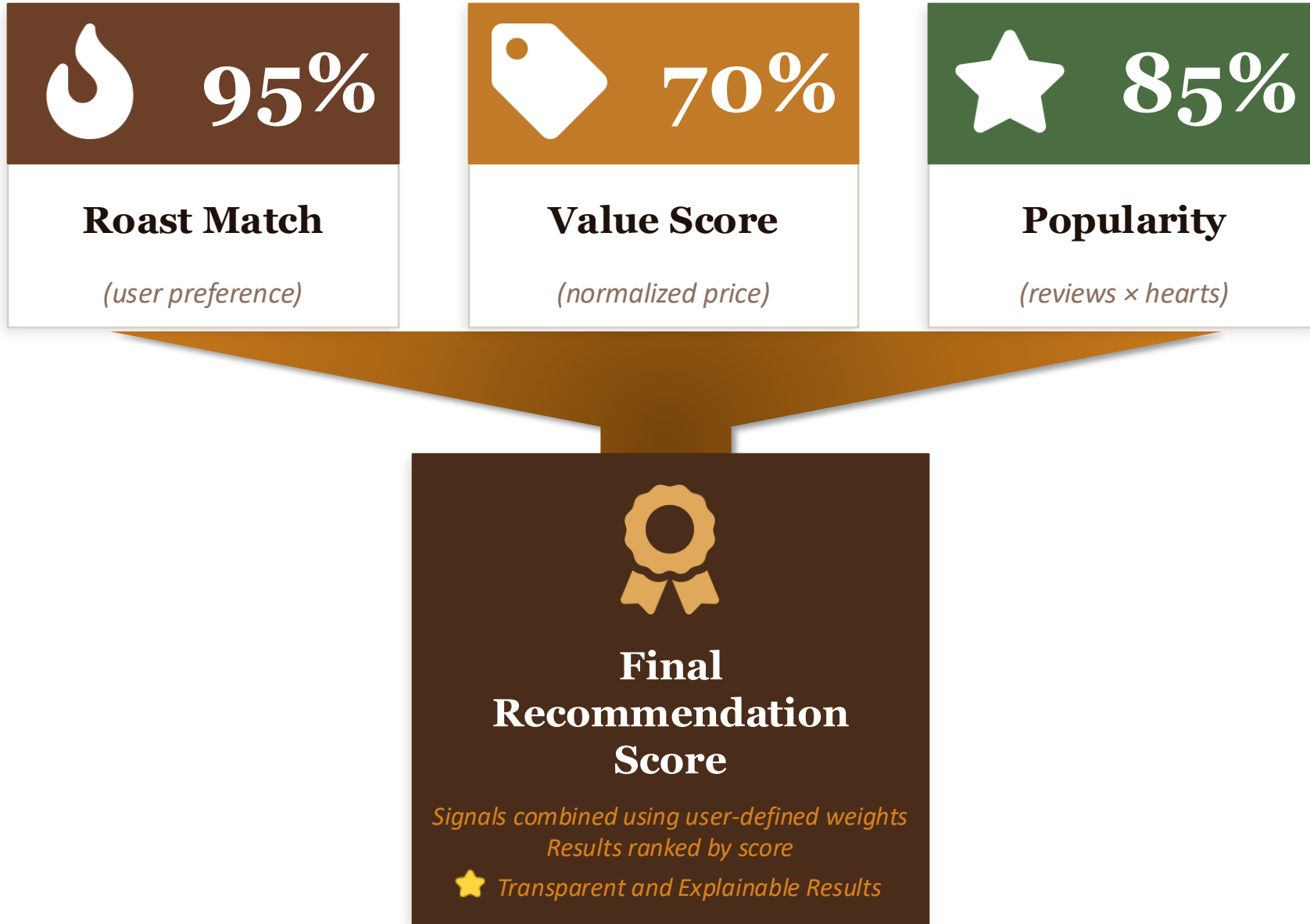
Large Bulk

Appears cheapest!



Solution: Reference bag selection + tier adjustment factor → normalized price per ounce

Recommendation Scoring



Recommendation Output Object

Recommendation Dataclass

Core Fields



product_name

Coffee product name



roaster

Roaster brand



roast_type

Light / Medium / Dark



café_location

Roaster cafe

Additional Data



available_sizes

Size + price list



tasting_notes

Aggregated review notes



review_texts

User reviews



match_reasons

Why recommended

Metadata



score

Final ranking score



heart_percentage

Positive review ratio



total_reviews

Review count



reference_price_per_oz

Normalized price

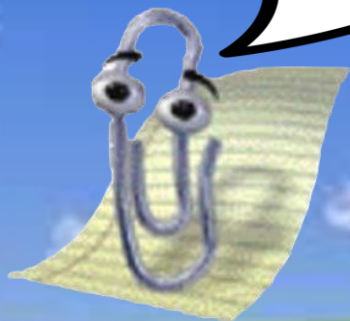
available_ground • blend • single_origin

UI / Demo

It looks like your users have a wide range of preferences.
Do you want to give them customization options?

Hmmm..

Sure!



♥ ♥ ♥ Match Maker Weights ♥ ♥ ♥

Importance of lower price (1 = least, 5 = most)

5

☹ ☹ ☹ Coffee Questions ☹ ☹ ☹

Are you only interested in single origin beans?

☒ Single origin only

☐ No preference



MATCH #1

Diablo

Dark Roast • Ladro Roasting
notes of brown sugar, chocolate, walnut

MATCH SCORE

93%

PRODUCTS

12oz (\$20.98), 80oz (\$92.13)

TRY BEFORE YOU BUY

CAFFE LADRO

[333 108TH AVE NE STE 175](#)



Demo

Coffee Match

Find the Washington Bean of your Dreams

☕ ☕ ☕ Coffee Questions ☕ ☕ ☕

Are you only interested in single origin beans?

- ☒ Single origin only
☐ No preference

Ground or whole beans (do you have a grinder)?

- ☒ Whole beans (yes)
☐ Pre-ground (no)

What's your roast preference?

- ☒ Light
☐ Medium
☐ Dark
☐ No preference / I'm not sure

Are you looking for a caffeinated or decaf coffee?

♥ ♥ ♥ Match Maker Weights ♥ ♥ ♥

Importance of lower price (1 = least, 5 = most)



Importance of product popularity (1 = least, 5 = most)



Importance of roast preference (1 = least, 5 = most)



Next Steps



Ability to add additional products



User login; ability to leave reviews



Product availability/tasting pipeline improvement

Questions?